

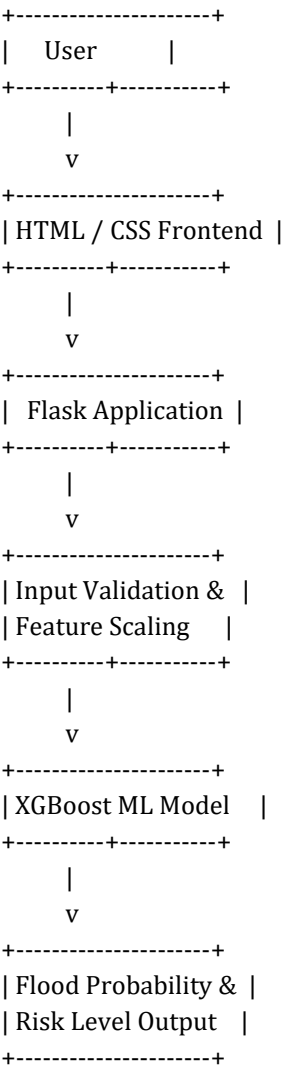
Solution Architecture

Project: Rising Water – AI-Based Flood Prediction System

1. Architecture Overview

The Rising Water system follows a layered architecture consisting of a user interface, Flask application, preprocessing layer, trained XGBoost model, and prediction output.

2. System Architecture Diagram



3. Components

Component	Purpose
Frontend	Collects 20 environmental inputs from the user.
Flask Backend	Receives requests and coordinates prediction.

Scaler	Normalizes input data before prediction.
XGBoost Model	Predicts flood probability.
Result Module	Displays probability and Low/Medium/High risk.

4. Data Flow

1. User enters environmental parameters.
2. Flask validates the inputs.
3. Scaler preprocesses the values.
4. XGBoost model predicts flood probability.
5. System displays probability and risk level.

5. Conclusion

The layered architecture makes the Rising Water application modular, maintainable, and suitable for future enhancements such as live weather APIs and GIS integration.